

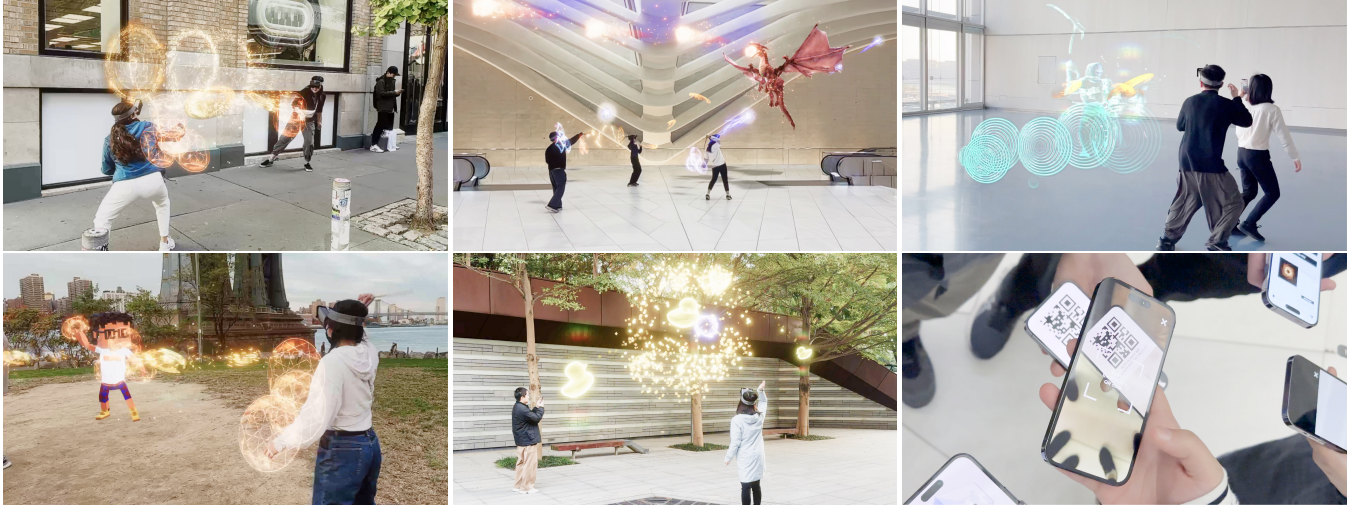
# MOFA: Multiplayer Omnipresent Fighting Arena

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**Figure 1:** Actual gameplay screenshots from a third-person spectator view of our five Spontaneous Collocated Mixed Reality (SCMR) game prototypes using the MOFA framework: (a) *The Duel* – A competitive play on the streets of SoHo, NYC; (b) *The Dragon* – A puppeteering play at the Oculus Transportation Hub, NYC; (c) *The Ghost* – An asymmetric play inside a museum in Shenzhen; (d) *The Training* – A single-player play in a park under the bridge, Brooklyn; (e) *The Duck* – A collaborative play in a common area between office buildings in Shenzhen; (f) The final image depicts the quick registration process for spontaneous collocation using InstantCopresence [Hu et al. 2023a].

## ABSTRACT

As Mixed Reality (MR) head-mounted displays (HMDs) become more widely used, the context of collocated MR gaming is shifting from controlled, private environments to spontaneous, public settings in-the-wild. This shift transforms everyday locations into potential gaming stages, blurring the line between designated play areas and the public sphere. We term this phenomenon Spontaneous Collocated Mixed Reality (SCMR). This transformation necessitates game designs that accommodate both HMD wearers ("Wizards") and non-HMD wearers ("Muggles"). To explore the design space of synchronous social bodily interplay in SCMR, we developed the Multiplayer Omnipresent Fighting Arena (MOFA) framework. This framework leverages device asymmetry and role diversification to actively engage all participants. Using this framework, we created five game prototypes—"The Duel", "The Dragon", "The Ghost",

"The Training", and "The Duck". These games, inspired by scenarios from fantastic fiction such as Harry Potter, Game of Thrones, and Ghostbusters, demonstrate that strategically involving "Muggles" enhances social engagement and acceptance. This helps us to speculate a shared mixed reality future that is more inclusive and entertaining in a ubiquitous and spontaneous manner.

## CCS CONCEPTS

• **Human-centered computing** → Interaction design theory, concepts and paradigms; **Mixed / augmented reality**; Collaborative and social computing systems and tools; Participatory design.

## KEYWORDS

Augmented reality, Collaborative mixed reality, Social bodily play, Bystander inclusion, Game design, Spontaneous Collocated Mixed Reality

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## 1 INTRODUCTION

In the emerging field of Spontaneous Collocated Mixed Reality (SCMR), designers face unique challenges and opportunities as the physical and digital worlds merge in the public sphere. The increasing prevalence of Mixed Reality (MR) Head-Mounted Displays (HMDs), like the Vision Pro and Meta Quest Pro 3, is turning everyday environments into spontaneous interaction and gaming arenas. This transformation resembles the blend of ordinary and extraordinary found in fantasy narratives like Harry Potter, where the boundaries between the mundane and fantastical are blurred.

The concept of collocated play in mixed reality is well-established [Auda et al. 2023; van Delden et al. 2018]. However, the shift to spontaneous, uncontrolled, in-the-wild, public environments requires not only technological infrastructures that allow swift establishment of collocated mixed reality sessions, but also game designs that understand the intricate social challenges in the shared physical space of HMD wearers ("Wizards") and non-HMD wearers ("Muggles") [Gonçalves et al. 2023]. This necessitates a flexible and inclusive design framework that takes into account the experiences of all - whether they are active participants with HMD devices, passersby with Handheld AR (HAR) technologies, or those without any devices.

To address this challenge, our work introduces the Multiplayer Omnipresent Fighting Arena (MOFA), a game framework designed not only for the swift establishment of spontaneous collocated multiplayer mixed reality sessions but also for easy configuration to various SCMR game prototypes, such as leveraging the inherent device asymmetry and role differentiation. Using the MOFA framework, we developed a series of SCMR game prototypes — "The Duel", "The Dragon", "The Ghost", "The Training", and "The Duck" — each embodying different facets of the MOFA framework [Hu et al. 2023b].

## 2 DEMO

### 2.1 Demo 1: MOFA - The Duel

Drawing inspiration from the Wizardry Duel in the Harry Potter series and dodgeball, we developed a game called 'The Duel'. The premise of the game is that players compete against each other by throwing virtual magic energy balls at one another while taking care to avoid being hit themselves. As it is not possible to cause physical harm to a real player within the MR environment, we created virtual shields in front of each player that can be destroyed by the magic energy balls. The goal of the game is to destroy as many shields as possible without allowing one's own shield to be destroyed within a time limit of 90 seconds. Those with handheld AR devices can spectate the game in a third-person perspective.

### 2.2 Demo 2: MOFA - The Dragon

In this demonstration, we explore a design strategy that offers HAR players a more active role beyond a third-person spectator. Drawing inspiration from a classic dragon-hunting myth, we assign one HAR player as a puppeteer to control agents like dragons. HAR provides players with efficient input methods and support for intricate interactions. This setup allows players to operate their agents easily, avoiding the inefficiencies or inaccuracies of hand gesture control.

The puppeteers' third-person view offers them freedom of perspective, enabling abilities that exceed physical constraints, such as flying. This puts the players in a situation that is far more powerful than themselves. The HMD opponents can also enjoy combat with the agent that they know is controlled by the HAR player, which provides a flow with uncertain behavior and intelligence behind the agent.

### 2.3 Demo 3: MOFA - The Ghost

This demonstration, inspired by the iconic film Ghostbusters, featured a game designed to explore a strategy for partial asymmetric information. This strategy could enhance collaboration among collocated multiplayer by leveraging their devices' capabilities. The game included three players: an HMD player known as the "ghostbuster", a HAR player known as the "detector", and another HAR player controlling the "ghost" known as the "ghost puppeteer". By default, the ghost was invisible to both HAR players. The "detector" could fire a detection wave to make the "ghost" visible, while the "ghostbuster", equipped with the HMD device, could throw an energy ball to harm the ghost. However, the "ghostbuster" could only see the ghost after the "detector" had successfully detected it for a few seconds. To defeat the enemy, players with different devices had to collaborate effectively and quickly before being eliminated.

## 3 CONCLUSION

In this work, we introduce the MOFA framework, a flexible and powerful tool for prototyping SCMR games. Utilizing local networking and spatial anchor sharing technologies as described in [Hu et al. 2023a], which do not require internet connections or Wi-Fi routers, we enable spontaneous MR collocated gaming scenarios in both private and public, in-the-wild environments. The framework allows game designers to leverage the physical affordances of AR or MR devices to assign distinct digital roles to players. This approach fosters deeper engagement and interaction among all participants, whether they have an HMD or not. Consequently, using the MOFA framework, we demonstrate five different prototyping games, inspired by fantastic fiction scenarios, which help us to speculate on a shared mixed reality future that is more inclusive and entertaining in a ubiquitous and spontaneous manner.

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